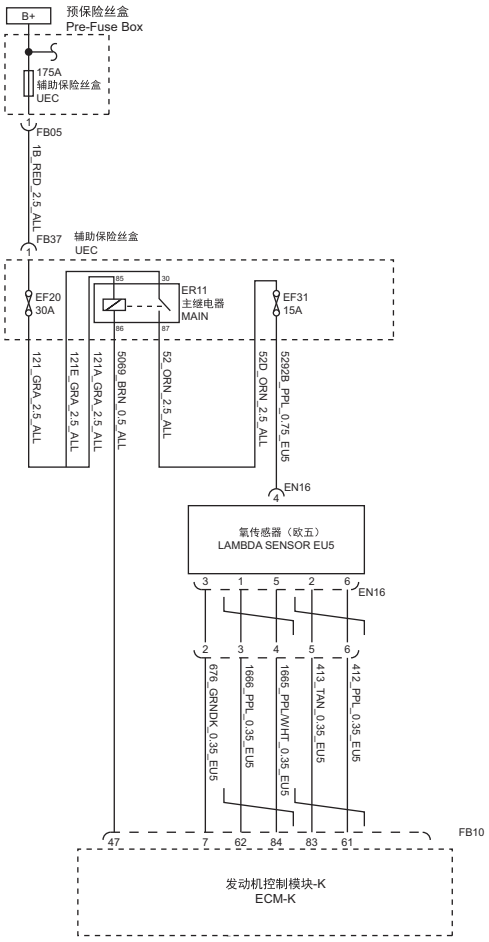
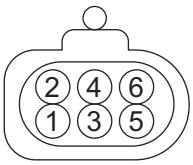
P0030

Description of fault diagnosis codes
P0030-11：SCG error of the LSU Heater Powerstage
P0030-12：SCB error of the LSU Heater Powerstage
P0030-29：Open Load error of the LSU Heater Powerstage
P0030-64：Over temperature error of the LSU Heater Powerstage
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Fault in harness or harness plug • Oxygen sensor fault
Description of circuit diagram
The heated oxygen sensor is used to monitor the fuel control. The heated oxygen sensor will compare the oxygen content in ambient air to that in the exhaust flow. The heated oxygen sensor must reach the work temperature so as to provide accurate voltage signal. The heating element inside the heated oxygen sensor will minimize the time of reaching work temperature required by the sensor. The voltage is supplied to the heater via a fuse from main relay circuit. Provide ground signal to the low-voltage control circuit of heating element via engine control module.

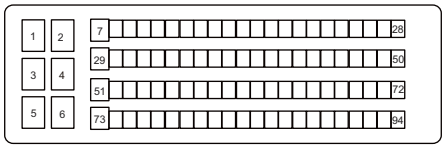
Crcuit diagram



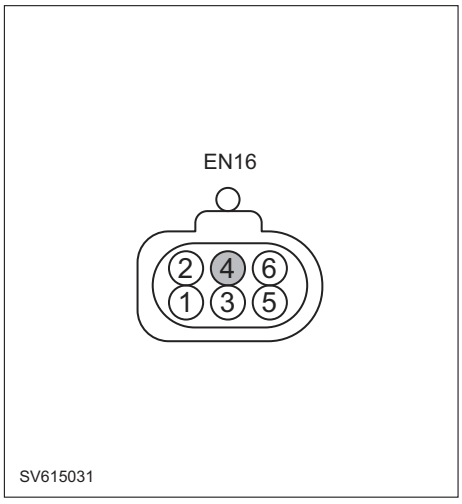
EN16

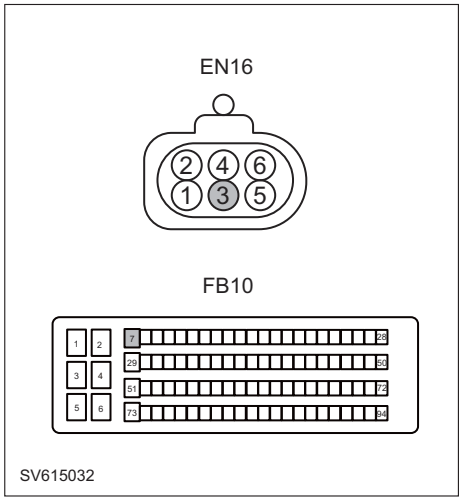
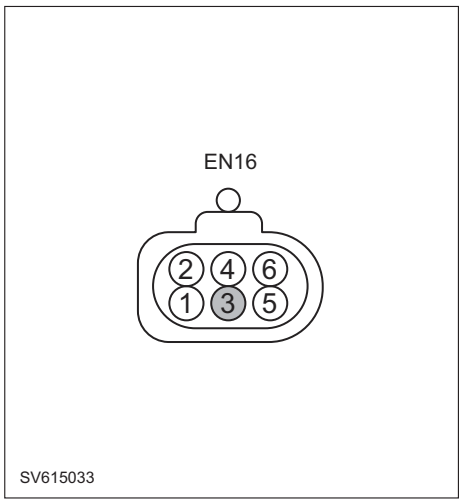


FB10



SV615022

<table>
<tr><td colspan="3">Troubleshooting procedures</td></tr>
<tr><td>Step 1</td><td colspan="2">General inspection.</td></tr>
<tr><td colspan="2"></td><td>1 Check the fuse EF31 for damage.
2 Check battery capacity.
3 Check the harness plugs of engine control module and oxygen sensor for damage, poor contact, aging and looseness.
4 Check whether the result is normal.</td></tr>
<tr><td colspan="2"></td><td>No ➞ Repair or replace the faulted parts.</td></tr>
<tr><td colspan="3">Yes ↓</td></tr>
<tr><td>Step 2</td><td colspan="2">Check fault codes.</td></tr>
<tr><td colspan="2"></td><td>1 Connect the fault diagnosis instrument to diagnosis interface.
2 Turn the ignition switch to ON position.
3 Diagnose the engine control system with diagnostic apparatus.
4 Are there any other fault codes except for P0030?</td></tr>
<tr><td colspan="2"></td><td>Yes ➞ Reference:DTC Summary List (EU5).</td></tr>
<tr><td colspan="3">No ↓</td></tr>
<tr><td>Step 3</td><td colspan="2">Check the power supply circuit of oxygen sensor.</td></tr>
<tr><td colspan="2"> 
</td><td>1 Turn the ignition switch to OFF.
2 Disconnect the harness plug of oxygen sensor.
3 Turn the ignition switch to ON.
4 Measure the voltage between the harness plug of oxygen sensor and ground.
Measuring circuit: the voltage between No.4 terminal of EN16 harness plug and ground.
Standard value: 10~14V
5 Check whether the result is normal?</td></tr>
<tr><td colspan="2"></td><td>No ➞ Repair or replace the faulted parts.</td></tr>
<tr><td colspan="3">Yes ↓</td></tr>
</table>

Step 4	Check the control signal of oxygen sensor heater (open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of engine control module. 3 Disconnect the harness plug of oxygen sensor. 4 Measure the resistance between the harness plug of oxygen sensor and that of engine control module. Measure circuit: The resistance between terminal 3 of EN16 plug and No.7 terminal of FB10 plug. **Standard value: ≈ 0Ω** 5 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 5	Check the control signal of oxygen sensor heater (short circuit).
	1 Check the resistance between the harness plug of oxygen sensor and ground. Measure circuit: The resistance between terminal 3 of EN16 plug and ground. **Standard value: ∞** 2 Check the voltage between the harness plug of oxygen sensor and ground. Measure circuit: The voltage between terminal 3 of EN16 plug and ground. **Standard value: 0V** 3 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 6	Check the oxygen sensor (parts inspection).
	1 Measure the resistance of terminal of oxygen sensor. Measure circuit: The resistance between terminal 3 and terminal 4 of oxygen sensor. **Standard value: 4.0~5.0V** 2 Check whether the result is normal?
	No → Replace the oxygen sensor.
Next ↓	
Step 7	Complete the check and confirm whether the system works normally nor not.